Supplementary Information

Task-wise model selection for heterogeneous ADMET prediction

Zhensheng Luo, Dachen Huang, Yanruisheng Shao, Qinze Yu and Yu Li

This supplement describes the datasets, model configurations and comparative analyses for the 22-task Therapeutics Data Commons (TDC) ADMET benchmark.

Contents

Dataset sources and endpoint definitions	2	Whole-task uncertainty: external models	10
Dataset coverage and applicability	3	Whole-task uncertainty: shared-pool AutoML	11
Supplementary Note 3A: data use and selection	4	Scale-free ablations and global configuration	12
Supplementary Note 3B: comparative experiments	5	Component-level OOF and official-test comparisons	13
Task-specific model configurations	6	Prediction-level AutoML comparison: graphical summary	14
External-model benchmark results	7	External-model comparison: graphical summary	15
Shared-pool comparison on all 22 tasks	8	Related-task supervision and target-only models	16
Baseline training and FLAML budget comparison	9	Molecular-subgroup results	17–20

List of supplementary tables and figures

Tables

Number	Title	Page
Table S1	Dataset sources, endpoint definitions and coverage	2–3
Table S2	Task-specific model components	6
Table S3	Trimole-Hybrid, MiniMol and Chemprop across 22 tasks (mean ± SD)	7
Table S4	Shared-pool comparison with two FLAML search budgets (mean ± population SD)	8
Table S4a	FLAML: 32 trials, 60 s, 120 s and 180 s (fixed nine-predictor pool)	9
Table S5	Chemprop configurations selected using seed 101	9
Table S6	Paired differences against MiniMol and Chemprop (95% intervals)	10
Table S7	Task-wise selection versus FLAML (120 s): paired differences on identical inputs	11
Table S8	Ablation changes and MLP late-fusion scores across 22 tasks	12
Table S9	Validation ranks for the global model-family choice	12
Table S10	Mean ranks of 27 methods across 20 development tasks	13
Table S11	Seven-method comparison on six official test tasks (five-run mean ± population SD)	13
Table S12	Source tasks for the original related-task models	16
Table S13	Results for the two target-only models	16
Table S14	Molecular-subgroup scores and paired differences across 22 tasks	17–20

Figures

Number	Title	Page
Figure S1	FLAML (120 s) and the Trimole-Hybrid reference across 22 tasks	14
Figure S2	Trimole-Hybrid, MiniMol and Chemprop across 22 tasks, grouped by official metric	15
Figure S3	Related-task and target-only model results against the archived public top-1 reference (CFA)	16

Overview of comparisons

Table S3 compares Trimole-Hybrid with MiniMol and Chemprop across 22 tasks, using target-only models for hepatocyte clearance and CYP3A4 substrate. Table S4 holds candidate predictions fixed; Table S4a examines search budgets in a separate nine-predictor pool. Tables S10–S11 report the component-level OOF study and the six-task evaluation. Note 3A describes data use, and Note 3B gives the comparison protocols.

Companion data. Run scores, model settings, row identifiers and analysis protocols are provided in the GitHub supplementary data.

Dataset sources and endpoint definitions

Supplementary Table S1. Dataset sources, endpoint definitions and coverage.

Task	Endpoint / unit	Source / identifier
Bioavailability	Human oral bioavailability; binary label	Ma et al. 2008 10.1016/j.jpba.2008.03.023
Caco-2	Caco-2 cell effective permeability; logPapp (permeability in cm/s)	Wang et al. 2016 10.1021/acs.jcim.5b00642
HIA	Human intestinal absorption; binary label	Hou et al. 2007 10.1021/ci600343x
Lipophilicity	Lipophilicity; experimental logD	AstraZeneca 2016; MoleculeNet 10.6019/CHEMBL3301361
P-gp	P-glycoprotein inh.; binary inhibitor label	Broccatelli et al. 2011 10.1021/jm101421d
BBB	Blood-brain barrier penetration; binary penetration label	Martins et al. 2012 10.1021/ci300124c
PPBR	Plasma protein binding rate; percent bound	AstraZeneca ChEMBL assay collection 10.6019/CHEMBL3301361
VDss	Steady-state volume of distribution; L/kg	Lombardo et al. 2016 10.1021/acs.jcim.6b00044
CYP2C9 sub.	CYP2C9 sub. status; binary sub. label	Carbon-Mangels and Hutter 2011; admetSAR 10.1002/minf.201100069
CYP2C9 inh.	CYP2C9 inh.; binary inhibitor label	Veith et al. 2009 10.1038/nbt.1581
CYP2D6 sub.	CYP2D6 sub. status; binary sub. label	Carbon-Mangels and Hutter 2011; admetSAR 10.1002/minf.201100069
CYP2D6 inh.	CYP2D6 inh.; binary inhibitor label	Veith et al. 2009 10.1038/nbt.1581
CYP3A4 sub.	CYP3A4 sub. status; binary sub. label	Carbon-Mangels and Hutter 2011; admetSAR 10.1002/minf.201100069
CYP3A4 inh.	CYP3A4 inh.; binary inhibitor label	Veith et al. 2009 10.1038/nbt.1581
Hepatocyte CL	Human hepatocyte intrinsic clearance; uL/min/10^6 cells	AstraZeneca 2016; ChEMBL 10.6019/CHEMBL3301361
Microsome CL	Human liver microsome intrinsic clearance; mL/min/g protein	AstraZeneca 2016; ChEMBL 10.6019/CHEMBL3301361
Half-life	Human elimination half-life; hours	Obach et al. 2008 10.1124/dmd.108.020479
AMES	Bacterial reverse-mutation activity; binary mutagenicity label	Xu et al. 2012 10.1021/ci300400a
DILI	Drug-induced liver injury; binary DILI label	Xu et al. 2015; US FDA NCTR 10.1021/acs.jcim.5b00238
hERG	hERG channel blockade; binary blocker label	Wang et al. 2016 10.1021/acs.molpharmaceut.6b00471
LD50	Rat acute oral toxicity; log(1/(mol/kg))	Zhu et al. 2009 10.1021/tx900189p
Solubility	Aqueous solubility; log mol/L	Sorkun et al. 2019 10.1038/s41597-019-0151-1

Sources and identifiers refer to the datasets used in the reported analyses.

Dataset coverage and applicability

Supplementary Table S1 (continued). Dataset sources, endpoint definitions and coverage.

Task	Train / valid / test	Unique	Scope / limitation
Bioavailability	448 / 64 / 128	640	Structural scaffold split does not represent formulation or population shifts
Caco-2	637 / 91 / 182	906	Applicable to the chemical and assay domain represented by the curated set
HIA	403 / 58 / 117	578	Does not model formulation or patient-level absorption variability
Lipophilicity	2940 / 420 / 840	4200	Restricted to the measured chemical space and assay conditions
P-gp	851 / 122 / 245	1212	Does not establish transport direction or clinical drug-drug interaction risk
BBB	1421 / 203 / 406	1975	Not a quantitative brain-exposure or transporter-specific endpoint
PPBR	1952 / 279 / 559	1797	Binding-rate measurements in the represented chemical space; row-level species assignments were not verified
VDss	791 / 113 / 226	1111	Does not replace population pharmacokinetic modeling
CYP2C9 sub.	467 / 67 / 135	666	CYP isoform overlap means cross-task molecular overlap must be filtered for transfer models
CYP2C9 inh.	8462 / 1209 / 2419	12090	Applicable to assay-defined inh. rather than in vivo metabolism
CYP2D6 sub.	465 / 67 / 135	664	CYP isoform overlap means cross-task molecular overlap must be filtered for transfer models
CYP2D6 inh.	9187 / 1313 / 2626	13126	Applicable to assay-defined inh. rather than in vivo metabolism
CYP3A4 sub.	468 / 67 / 135	667	CYP isoform overlap means cross-task molecular overlap must be filtered for transfer models
CYP3A4 inh.	8626 / 1233 / 2467	12326	Applicable to assay-defined inh. rather than in vivo metabolism
Hepatocyte CL	848 / 122 / 243	1020	Source-task overlap must be audited before cross-assay transfer; target-only primary analysis uses no source-task rows
Microsome CL	770 / 111 / 221	1102	Evaluation is limited to the chemical space and intrinsic-clearance assay conditions represented by this benchmark
Half-life	465 / 67 / 135	665	Does not transfer directly to unrepresented populations or dosing regimens
AMES	4684 / 1137 / 1457	7255	Bacterial mutagenicity does not represent all genotoxicity or human toxicity
DILI	331 / 48 / 96	475	Small benchmark does not represent the full mechanistic spectrum of liver injury
hERG	457 / 66 / 132	648	Blockade labels do not directly quantify clinical arrhythmia risk
LD50	3970 / 1937 / 1478	7342	Rat acute oral toxicity is not a general human-toxicity endpoint
Solubility	5045 / 2940 / 1995†	9980	Predictions should not be transferred across unrepresented formulation conditions

Rows are records in the analysis files; unique SMILES counts distinct strings before molecular standardization. The test column gives scored rows. † For solubility, the official test set contains 1,997 rows and the matched analysis uses 1,995 (Note 3A). Validation fractions follow the saved task-specific partitions. Class prevalence and measurement limitations are in the dataset provenance record in the GitHub supplementary data. Applicability is described by dataset chemistry and assay scope; no formal model-specific applicability domain was estimated.

Supplementary Note 3A: data use and selection

Task-specific model configurations

The supplied training and validation partitions were used during model development, and the supplied test partition was used only for final scoring. Each endpoint then uses an individual predictor or a prediction combination (Table S2). Candidates were ranked on validation performance or internal scaffold cross-validation. The control methods in Note 3B are fitted and selected on development data only.

Feature extraction and base fitting. Fingerprints and descriptors were computed with fixed settings. For the original configurations, pretrained ChemBERTa and KPGT encoders were kept frozen. Supervised EPT, prediction heads and chemical-feature models used training rows, with validation-based early stopping where applicable. Target-only model training is described in Table S13. C and K denote ChemBERTa and KPGT; E/EPT is the supervised geometry-sensitive encoder branch. FP denotes chemical fingerprints, XL expanded chemical features, and AP the average-precision-surrogate models used in the CYP2D6 combinations.

Hyperparameters and candidate selection. BBB used five-seed validation comparisons, and six chemical-feature endpoints used internal scaffold cross-validation. Other task-specific searches and fixed settings are listed in the companion configurations.

Blend weights and calibration. Six original combinations used validation-selected weights; the others used fixed or equal weights. Target-only blends are specified in Table S13. For the shared-pool comparison, rank and z-score transformations operate on prediction vectors without endpoint labels. No separate calibration step was used.

Final refitting. Train-plus-validation refitting followed candidate ranking where it was used. Other configurations retained fitted fold models or combined their saved predictions. Table S2 and the companion settings identify components, transformations, weights and seed sets.

Development data and local selection

For task t, development-data membership is

$$\mathcal{D}_{\mathrm{dev},t} = \mathcal{D}_{\mathrm{train},t} \cup \mathcal{D}_{\mathrm{valid},t}, \quad \mathcal{D}_{\mathrm{train},t} \cap \mathcal{D}_{\mathrm{valid},t} = \varnothing. \tag{S1}$$

Equation (S1) specifies which rows belong to development data; the fitting rows depend on the stage. Local configuration selection is written as

$$\theta^*_{t,a} = \arg\max_{\theta \in \Theta_{t,a}} Q_{t,a}(\theta), \tag{S2}$$

where $Q_{t,a}$ is the validation or internal-CV utility recorded for branch or method a. MAE is sign-reversed so that every metric is maximized. Q uses development data only; the selection basis for each final configuration is given under Task-specific model configurations.

Solubility test coverage

All three models are compared on the same 1,995 solubility test rows. Trimole-Hybrid saved predictions for 1,995 of the 1,997 official rows; the two omissions contained pyridine N-oxide SMILES with invalid nitrogen-valence representations. MiniMol and Chemprop processed identity-corrected strings for these molecules. Their existing predictions were restricted to the matched rows, without changing models or parameters. Trimole-Hybrid used 5,045 training rows and 2,940 validation rows (7,985 in total); these fitting rows were unchanged. Table S3 reports matched five-run MAEs and population SDs. The companion data also contain the original 1,997-row baseline results and the structure-correction map.

Supplementary Note 3B: comparative experiments

22-task shared-pool comparison (Table S4)

Each task/run uses one aligned candidate-prediction matrix. Task-specific components are supplemented by reference candidates on matching development/test rows; target-only tasks use their own components. Recorded transformations and weights reconstruct the task-wise configuration, with zero weight on added columns. Rebuilt components can produce scores different from the main table. Table S4 reports seven primary methods, including FLAML at 120 s, plus its 180-s budget setting.

Control methods rank candidates or fit combination models on development predictions. Tuned stacking and FLAML use three folds grouped by largest-fragment molecular connectivity, with scalers fitted within each training fold. Cross-validation refits the combination layer while the base predictions stay fixed. Test structures identify and align evaluation rows; the fitting worker receives development files, and a separate evaluator scores fixed outputs for each comparison arm. The task-wise arm keeps the 20 original configuration choices (Table S2); the control methods are fitted and selected on development data. Table S4a uses a separate fixed nine-predictor pool.

20-task controlled study (S10)

We split target-task development data by molecular scaffold/connectivity components into base-training E, combination-fitting M, setting-selection V and internal evaluation H. The study uses no official TDC test rows or labels; H is a holdout within the development data. Fingerprints and descriptors were fixed, and ChemBERTa/KPGT encoders stayed frozen. Supervised EPT was fitted on E_{fit}, with early stopping on E_{stop}. Heads, scalers and target transformations used their designated fitting rows; no separate calibration step was used.

Primary combinations were fitted on M, and their settings were selected on V. For component-level out-of-fold (OOF) methods, we refitted EPT and heads in three component folds of E_{fit}, excluding the predicted component from fitting and early stopping. These predictions supplemented M during meta-training. After selection, the combination layer was refitted on $M \cup V$, including the extra OOF rows for OOF methods. The original E-trained predictors generated the same candidate matrix on H for every method; fold-specific fits supplied additional meta-training rows and computation.

B selects one candidate by its pooled $M \cup V$ score; pooled top-3 uses the same ranking. U averages all nine candidates, and O uses grouped mean-utility top-3 ranking. L is an adapted three-candidate blending rule: it selects candidates on M and searches 231 simplex weights at steps of 0.05 on V. F fits convex weights with an anchor and relative-loss penalty. The complete method definitions are given with Table S10. Eligibility checks used development labels, including H, for class support and grouping feasibility; H performance was reserved for evaluation. Table S2 lists the main benchmark configurations.

Six-task and target-only evaluations

The six-task comparison (Table S11) used seven methods and complete five-run predictions. Methods and tasks were fixed before scoring. Development components sharing standardized connectivity or a nonempty Murcko scaffold with official test molecules were excluded before fitting. Models followed the $E/M/V$ training scheme, with internal H unused, and official labels were read after predictions were fixed. The separate target-only evaluations (Table S13) used target-task development validation followed by full-development refitting. Both evaluations reuse the official test partitions of the original benchmark and are therefore not independent test sets.

Companion data and reproducibility

The GitHub supplementary data contain scores, settings, row identifiers, protocols and analyses, including both solubility row sets. The index links source records to Tables S1–S14 and Figures S1–S3, and identifies the component matrices for the main table and original ablations.

Code and analysis archive (version 2026.10.24): https://doi.org/10.5281/zenodo.23602309. Repository: https://github.com/dchen0212/trimole_hybrid. The archive contains the original code and saved predictions, a result-to-code map, dependency instructions and a file-hash manifest. The 1,995-row rescoring records are available in the GitHub supplementary data.

Task-specific model configurations

Supplementary Table S2. Task-specific model components.

Task	Metric	Model / combination	Chem	Seq	Graph	3D	Blend	Bag
TDC.Bioav	AUROC	Fixed chem ExtraTrees	•		•			•
TDC.Caco2	MAE	Fixed chem ExtraTrees	•	•	•			•
TDC.HIA	AUROC	Task-specific chem blend	•				•	•
TDC.Lipo	MAE	XL/chem hybrid blend	•	•	•	•	•	•
TDC.Pgp	AUROC	KPGT/EPT logit blend		•	•	•	•	
TDC.AqSol	MAE	FP-XGB/KPGT-EPT/XL blend	•	•	•	•	•	
TDC.BBB	AUROC	Descriptor ExtraTrees	•					
TDC.PPBR	MAE	KPGT/fingerprint blend	•		•		•	•
TDC.VD	Spearman	KPGT/EPT rank blend	•		•	•	•	•
TDC.CYP2C9-S	AUPRC	Fixed chem XGBoost	•	•	•	•		•
TDC.CYP2C9-I	AUPRC	Fixed chem XGBoost	•		•	•		•
TDC.CYP2D6-S	AUPRC	AP/XL/descriptor blend	•	•		•	•	•
TDC.CYP2D6-I	AUPRC	KPGT/EPT/FP rank blend	•		•	•	•	•
TDC.CYP3A4-S	AUROC	Target-only C/K RBF-SVM blend		•	•		•	
TDC.CYP3A4-I	AUPRC	XL/KPGT-EPT logit blend	•	•	•	•	•	•
TDC.CL-Hepa	Spearman	Target-only C/K/KPGT/Uni-Mol blend		•	•	•	•	
TDC.CL-Micro	Spearman	KPGT/EPT seedbag			•	•	•	•
TDC.Half_Life	Spearman	C+K descriptor/z-score blend	•	•	•		•	•
TDC.AMES	AUROC	Repeated fingerprint XGBoost	•					•
TDC.DILI	AUROC	XL/KPGT-EPT raw blend	•	•	•	•	•	•
TDC.hERG	AUROC	Fixed chem ExtraTrees	•	•		•		•
TDC.LD50	MAE	Fixed chem XGBoost	•	•	•	•		•

Filled entries mark components. Chem: fingerprints/descriptors; Seq: SMILES features; Graph: KPGT features; 3D: EPT, or Uni-Mol for target-only hepatocyte clearance; Blend: prediction combination; Bag: seed/fold averaging. CL-Hepa and CL-Micro denote hepatocyte and microsome clearance. The CYP3A4 substrate and hepatocyte rows use the target-only models in Table S13. Original related-task models supply the ablation records in Table S8 and the comparisons in Table S13. Data-use notation is defined in Note 3A; component identifiers, transformations, weights and seeds are in the companion files.

Molecule-level sensitivity analyses (main Figure 4)

AMES test_297 was selected after inspecting its label and prediction. A reconstructed fingerprint-XGBoost model scored both the original and edited structures; the stored original predictor was not exactly replayable for arbitrary structures. Deutivacaftor is an external example evaluated by the exact P-gp-inhibition pipeline, without a matched experimental inhibition label. Its ring carbonyl was replaced by an exocyclic methylene. These analyses quantify model-output sensitivity to the specified edits.

Case-study records. The AMES replay audit is listed under manuscript Figure 4 in the GitHub supplementary data index.

External-model benchmark results

Supplementary Table S3. Trimole-Hybrid, MiniMol and Chemprop across 22 tasks (mean ± SD).

Task	Metric	Trimole-Hybrid	MiniMol	Chemprop
Bioavailability	AUROC	0.7222 ± 0.0040	0.6967 ± 0.0182	0.6002 ± 0.0214
Caco-2	MAE	0.2836 ± 0.0023	0.3582 ± 0.0201	0.3339 ± 0.0189
HIA	AUROC	0.9943 ± 0.0017	0.9951 ± 0.0009	0.9818 ± 0.0134
Lipophilicity	MAE	0.4729 ± 0.0065	0.4483 ± 0.0051	0.4328 ± 0.0077
P-gp	AUROC	0.9384 ± 0.0036	0.9442 ± 0.0026	0.9046 ± 0.0053
BBB	AUROC	0.9278 ± 0.0005	0.9235 ± 0.0033	0.8863 ± 0.0094
PPBR	MAE	7.3306 ± 0.0986	7.8774 ± 0.0772	7.8901 ± 0.1297
VDss	Spearman	0.6610 ± 0.0047	0.5592 ± 0.0191	0.5339 ± 0.0254
CYP2C9 sub.	AUPRC	0.4434 ± 0.0234	0.4699 ± 0.0242	0.4004 ± 0.0335
CYP2C9 inh.	AUPRC	0.7906 ± 0.0026	0.8274 ± 0.0032	0.7644 ± 0.0082
CYP2D6 sub.	AUPRC	0.7475 ± 0.0096	0.7046 ± 0.0265	0.6081 ± 0.0481
CYP2D6 inh.	AUPRC	0.7192 ± 0.0055	0.7213 ± 0.0033	0.6728 ± 0.0061
CYP3A4 sub.	AUROC	0.6736 ± 0.0011	0.6471 ± 0.0108	0.5986 ± 0.0252
CYP3A4 inh.	AUPRC	0.8841 ± 0.0032	0.8783 ± 0.0009	0.8684 ± 0.0053
Hepatocyte CL	Spearman	0.5378 ± 0.0058	0.4674 ± 0.0157	0.2926 ± 0.0508
Microsome CL	Spearman	0.6428 ± 0.0058	0.6258 ± 0.0058	0.5369 ± 0.0442
Half-life	Spearman	0.4914 ± 0.0098	0.5108 ± 0.0123	0.4482 ± 0.1549
AMES	AUROC	0.8719 ± 0.0006	0.8562 ± 0.0040	0.8236 ± 0.0059
DILI	AUROC	0.9086 ± 0.0055	0.9501 ± 0.0098	0.8908 ± 0.0097
hERG	AUROC	0.8634 ± 0.0023	0.8432 ± 0.0161	0.8023 ± 0.0156
LD50	MAE	0.6031 ± 0.0038	0.5887 ± 0.0108	0.6116 ± 0.0114
Solubility	MAE	0.7384 ± 0.0005	0.7391 ± 0.0099	0.7649 ± 0.0149

Lower MAE and higher AUROC, AUPRC and Spearman indicate better performance. Trimole-Hybrid uses 20 original endpoints and two target-only models. Baseline training is specified on page 9 and in Table S5. All three methods use the same 1,995 solubility test rows. Values are five-run means and SDs; baseline and target-only SDs use the population convention. Table S6 reports paired intervals, and Table S4 compares combination methods on shared inputs.

Shared-pool comparison on all 22 tasks

Supplementary Table S4. Shared-pool comparison with two FLAML search budgets (mean ± population SD).

Task	Metric	Task-wise selection	Single	Uniform	Top-3	Stacking	Tuned stacking	FLAML 120 s	FLAML 180 s
Bioavailability	AUROC	0.7228 ± 0.0028	0.6694 ± 0.0100	0.7113 ± 0.0085	0.6884 ± 0.0181	0.6017 ± 0.0453	0.6594 ± 0.0341	0.6593 ± 0.0207	0.6734 ± 0.0126
Caco-2	MAE	0.2824 ± 0.0016	0.3197 ± 0.0753	0.2893 ± 0.0012	0.3030 ± 0.0166	0.3638 ± 0.0218	0.3362 ± 0.0113	0.3378 ± 0.0087	0.3396 ± 0.0070
HIA	AUROC	0.9940 ± 0.0013	0.9799 ± 0.0163	0.9838 ± 0.0039	0.9865 ± 0.0004	0.9855 ± 0.0053	0.9857 ± 0.0055	0.9834 ± 0.0129	0.9834 ± 0.0129
Lipophilicity	MAE	0.4729 ± 0.0058	0.4902 ± 0.0000	0.5041 ± 0.0017	0.4725 ± 0.0048	0.4682 ± 0.0037	0.4687 ± 0.0039	0.4687 ± 0.0043	0.4686 ± 0.0044
P-gp	AUROC	0.9384 ± 0.0036	0.9237 ± 0.0129	0.9410 ± 0.0033	0.9361 ± 0.0029	0.9295 ± 0.0081	0.9430 ± 0.0037	0.9304 ± 0.0085	0.9303 ± 0.0079
BBB	AUROC	0.9264 ± 0.0013	0.9072 ± 0.0056	0.9155 ± 0.0041	0.9245 ± 0.0041	0.8209 ± 0.0137	0.8807 ± 0.0324	0.8862 ± 0.0349	0.8877 ± 0.0334
PPBR	MAE	7.3306 ± 0.0882	7.4934 ± 0.1446	8.3927 ± 0.0074	7.6539 ± 0.0521	8.8859 ± 0.1316	8.8561 ± 0.3613	8.6949 ± 0.3938	8.8100 ± 0.7900
VDss	Spearman	0.6610 ± 0.0042	0.6587 ± 0.0041	0.4210 ± 0.0050	0.6046 ± 0.0181	0.4117 ± 0.0191	0.4621 ± 0.0123	0.4954 ± 0.0722	0.4730 ± 0.1222
CYP2C9 sub.	AUPRC	0.4396 ± 0.0164	0.4638 ± 0.0316	0.4212 ± 0.0083	0.4453 ± 0.0231	0.4124 ± 0.0280	0.4524 ± 0.0036	0.3837 ± 0.0528	0.3837 ± 0.0528
CYP2C9 inh.	AUPRC	0.7898 ± 0.0015	0.7690 ± 0.0000	0.7863 ± 0.0007	0.7719 ± 0.0000	0.7824 ± 0.0030	0.7836 ± 0.0022	0.7751 ± 0.0075	0.7747 ± 0.0075
CYP2D6 sub.	AUPRC	0.7475 ± 0.0086	0.6913 ± 0.0336	0.7099 ± 0.0105	0.7374 ± 0.0198	0.5661 ± 0.0662	0.6938 ± 0.0414	0.6492 ± 0.0217	0.6692 ± 0.0480
CYP2D6 inh.	AUPRC	0.7192 ± 0.0049	0.6956 ± 0.0000	0.7245 ± 0.0029	0.7062 ± 0.0040	0.7258 ± 0.0016	0.7265 ± 0.0024	0.7235 ± 0.0047	0.7245 ± 0.0044
CYP3A4 sub.	AUROC	0.6735 ± 0.0011	0.6747 ± 0.0018	0.6722 ± 0.0006	0.6781 ± 0.0011	0.6797 ± 0.0010	0.6803 ± 0.0017	0.6743 ± 0.0101	0.6742 ± 0.0101
CYP3A4 inh.	AUPRC	0.8841 ± 0.0029	0.8743 ± 0.0000	0.8856 ± 0.0021	0.8803 ± 0.0033	0.8851 ± 0.0008	0.8859 ± 0.0012	0.8839 ± 0.0030	0.8839 ± 0.0028
Hepatocyte CL	Spearman	0.5379 ± 0.0059	0.5295 ± 0.0046	0.5363 ± 0.0046	0.5270 ± 0.0075	0.5312 ± 0.0036	0.5352 ± 0.0043	0.5279 ± 0.0081	0.5273 ± 0.0096
Microsome CL	Spearman	0.6428 ± 0.0052	0.5576 ± 0.0981	0.6469 ± 0.0041	0.5967 ± 0.0329	0.4752 ± 0.0308	0.5454 ± 0.0713	0.5611 ± 0.0145	0.5561 ± 0.0180
Half-life	Spearman	0.4914 ± 0.0088	0.4537 ± 0.0000	0.4247 ± 0.0095	0.4771 ± 0.0217	0.1341 ± 0.0225	0.2269 ± 0.1100	0.3764 ± 0.0571	0.3970 ± 0.0498
AMES	AUROC	0.8719 ± 0.0006	0.8713 ± 0.0021	0.8594 ± 0.0014	0.8720 ± 0.0013	0.8581 ± 0.0049	0.8657 ± 0.0072	0.8633 ± 0.0069	0.8617 ± 0.0055
DILI	AUROC	0.9086 ± 0.0049	0.8787 ± 0.0000	0.9105 ± 0.0094	0.9146 ± 0.0057	0.9077 ± 0.0101	0.9156 ± 0.0099	0.8869 ± 0.0129	0.8776 ± 0.0188
hERG	AUROC	0.8638 ± 0.0014	0.7608 ± 0.0000	0.8455 ± 0.0076	0.8216 ± 0.0110	0.7922 ± 0.0157	0.8346 ± 0.0166	0.8056 ± 0.0182	0.8003 ± 0.0208
LD50	MAE	0.6032 ± 0.0026	0.6044 ± 0.0049	0.5964 ± 0.0015	0.5950 ± 0.0057	0.6084 ± 0.0062	0.6071 ± 0.0063	0.6153 ± 0.0071	0.6142 ± 0.0065
Solubility	MAE	0.7384 ± 0.0005	0.7716 ± 0.0000	0.7544 ± 0.0046	0.7334 ± 0.0091	0.8722 ± 0.0080	0.8558 ± 0.0095	0.8209 ± 0.0217	0.8228 ± 0.0131

Lower MAE and higher other metrics indicate better performance. Every arm uses the same candidate matrix. The reconstructed task-wise arm differs from the main-table reference where its components were rebuilt. Single fitted prediction vectors are reused across run indices where applicable; other components vary by seed. Solubility uses 1,995 rows. Better-mean counts against the seven control columns are 20, 16, 15, 18, 15, 19 and 19. The 180-s FLAML means improve on 7 tasks, tie on 2 and worsen on 13 relative to 120 s. Counts use unrounded means. Costs cover combination-layer search and refitting. The task-wise arm keeps the 20 original configuration choices (Table S2); the control methods are fitted and selected on development data.

Shared-pool controls (S4). Fixed stacking uses logistic $C = 1$ for classification or Ridge $\alpha = 1$ for regression; tuned stacking searches nine C/α values from 10^{-4} to 10^{4}. Single-candidate and top-3 controls rank development scores, with candidate ID resolving ties. Averaging uses percentile ranks for Spearman and raw predictions for other metrics; the task-wise configuration retains its recorded transforms. FLAML 2.7.0 searches RF, ExtraTrees, XGBoost, LightGBM, CatBoost and scaled logistic regression/Ridge using the task-native metric. The primary 120-s and budget-check 180-s searches use identical inputs, folds, seeds and learners, one CPU thread, no subsampling and no internal ensembling. At 180 s, median actual search time was 180.007 s (179.743–227.663 s). Refit time is separate. Run-level records and source bindings accompany the comparison data.

Baseline training and FLAML budget comparison

MiniMol. We followed the pinned public implementation with a frozen encoder. Each of five repetitions averaged five heads trained for 25 epochs with batch size 32. Validation BCE/MSE selected checkpoints. Cantor-paired seeds specified scaffold splits; fit records document training/evaluation modes, random-number handling and compatibility adjustments.

Chemprop 2.3.1 (S5). We trained directed bond-message-passing networks from scratch, attempting twenty configurations on the seed-101 development split (successful-fit counts in S5). Search covered depth 2–6, hidden widths 300/600/900/1200, one/two FFN layers, dropout 0–0.4 and maximum learning rate 10^{-4}–10^{-2}. The selected configuration used seeds 101, 202, 303, 404 and 505, batch size 64, up to 50 epochs and validation patience 20, without full-development refitting. Validation loss resolved HIA validation-AUROC ties; half-life used the validation-selected log-target model. All five runs are reported.

Supplementary Table S5. Chemprop configurations selected using seed 101.

Task	Fits	Depth	Message	FFN	Layers	Dropout	Max LR
Bioavailability	20	6	600	1200	1	0.3	0.00449
Caco-2	20	2	600	1200	2	0.2	0.00444
HIA	20	3	1200	1200	1	0.4	0.00674
Lipophilicity	20	2	600	1200	2	0.2	0.00444
P-gp	20	6	300	1200	2	0.2	0.00354
BBB	20	3	1200	1200	1	0.4	0.00674
PPBR	20	2	1200	1200	1	0.0	0.00237
VDss	20	2	600	1200	2	0.2	0.00444
CYP2C9 sub.	20	6	600	1200	1	0.3	0.00449
CYP2C9 inh.	20	6	600	1200	1	0.3	0.00449
CYP2D6 sub.	20	4	600	1200	2	0.1	0.00059
CYP2D6 inh.	20	3	1200	1200	1	0.4	0.00674
CYP3A4 sub.	20	5	1200	300	1	0.3	0.00196
CYP3A4 inh.	20	4	600	1200	2	0.1	0.00059
Hepatocyte CL	20	6	600	1200	1	0.3	0.00449
Microsome CL	20	3	1200	1200	1	0.4	0.00674
Half-life	20	4	600	1200	2	0.1	0.00059
AMES	20	2	600	600	2	0.0	0.00409
DILI	20	6	300	1200	2	0.2	0.00354
hERG	19	5	1200	300	1	0.3	0.00196
LD50	20	2	600	1200	2	0.2	0.00444
Solubility	20	2	900	300	2	0.3	0.00127

Fits: successful trials out of 20. Message/FFN: hidden widths; Layers: FFN layer count. Initial/final LR is 0.1 times Max LR. Failed trials remain in the fit records.

Supplementary Table S4a. FLAML under four search settings in the fixed nine-predictor pool.

Five-run task means ± population SD. Each setting covers 22 tasks and 110 runs.

Task	Metric	32 trials	60 s	120 s	180 s
Bioavailability	AUROC	0.6603 ± 0.0207	0.6392 ± 0.0434	0.6529 ± 0.0548	0.6284 ± 0.0441
Caco-2	MAE	0.3409 ± 0.0207	0.3281 ± 0.0064	0.3363 ± 0.0271	0.3363 ± 0.0271
HIA	AUROC	0.9610 ± 0.0144	0.9649 ± 0.0083	0.9736 ± 0.0090	0.9736 ± 0.0090
Lipophilicity	MAE	0.5424 ± 0.0147	0.5349 ± 0.0033	0.5178 ± 0.0040	0.5178 ± 0.0040
P-gp	AUROC	0.9034 ± 0.0355	0.9053 ± 0.0283	0.9009 ± 0.0300	0.9183 ± 0.0179
BBB	AUROC	0.8914 ± 0.0190	0.8780 ± 0.0249	0.8780 ± 0.0249	0.8791 ± 0.0260
PPBR	MAE	9.2925 ± 0.4970	9.3019 ± 0.3524	9.9005 ± 0.2783	9.9005 ± 0.2783
VDss	Spearman	0.2790 ± 0.1652	0.3794 ± 0.0511	0.3591 ± 0.0391	0.3654 ± 0.0460
CYP2C9 sub.	AUPRC	0.3817 ± 0.0248	0.3584 ± 0.0355	0.3614 ± 0.0368	0.3614 ± 0.0368
CYP2C9 inh.	AUPRC	0.7650 ± 0.0139	0.7764 ± 0.0075	0.7746 ± 0.0117	0.7746 ± 0.0120
CYP2D6 sub.	AUPRC	0.6020 ± 0.0286	0.5656 ± 0.0719	0.5634 ± 0.0749	0.5553 ± 0.0767
CYP2D6 inh.	AUPRC	0.6966 ± 0.0147	0.7012 ± 0.0093	0.7064 ± 0.0031	0.7060 ± 0.0023
CYP3A4 sub.	AUROC	0.6349 ± 0.0444	0.6322 ± 0.0353	0.6304 ± 0.0347	0.6257 ± 0.0315
CYP3A4 inh.	AUPRC	0.8711 ± 0.0039	0.8710 ± 0.0027	0.8724 ± 0.0020	0.8707 ± 0.0014
Hepatocyte CL	Spearman	0.3879 ± 0.0794	0.4454 ± 0.0248	0.4614 ± 0.0153	0.4256 ± 0.0543
Microsome CL	Spearman	0.5225 ± 0.0177	0.5014 ± 0.0244	0.5143 ± 0.0205	0.5107 ± 0.0218
Half-life	Spearman	0.2390 ± 0.1024	0.1958 ± 0.0811	0.1936 ± 0.0665	0.1791 ± 0.0796
AMES	AUROC	0.8467 ± 0.0109	0.8492 ± 0.0076	0.8494 ± 0.0074	0.8501 ± 0.0071
DILI	AUROC	0.8728 ± 0.0211	0.8873 ± 0.0193	0.8823 ± 0.0168	0.8818 ± 0.0137
hERG	AUROC	0.7780 ± 0.0207	0.7773 ± 0.0173	0.7773 ± 0.0239	0.7844 ± 0.0248
LD50	MAE	0.6343 ± 0.0045	0.6333 ± 0.0048	0.6308 ± 0.0071	0.6300 ± 0.0067
Solubility	MAE	0.8419 ± 0.0157	0.8212 ± 0.0041	0.8224 ± 0.0038	0.8231 ± 0.0028

Setting	Learner families/task	Median actual search (s)	Better/tied/worse vs preceding	Main table better
32-trial cap	3	1.82	—	22/22
60 s	3	60.00	12/0/10	22/22
120 s	6	120.00	10/1/11	22/22
180 s	6	180.00	8/4/10	22/22

The 32-trial setting caps configuration count. The 32-trial/60-s settings use RF/ExtraTrees/XGBoost; 120/180 s use the expanded learner list with S4. Inputs, folds, seeds and learners stay fixed between 120 and 180 s.

Counts use unrounded means; displayed entries have four decimals. In this separate nine-predictor analysis, main-table means are better on all 22 tasks at every setting. Solubility uses 1,995 matched rows. The task-specific candidate-pool comparison is reported in Table S4.

Search times exclude refitting and upstream model fitting. Prediction rows were aligned by their order in the source files; run records and source identifiers are included with the data.

Whole-task uncertainty: external models

We used 2,000 paired scaffold-cluster bootstrap resamples, drawing the same rows for all methods within each saved run. Percentile 95% intervals quantify sampling uncertainty conditional on the fitted models and are unadjusted for multiple comparisons. Positive differences favor Trimole-Hybrid, with the sign reversed for MAE. Intervals require at least 30 clusters, five observations per class for classification and 1,900 valid resamples; subgroup intervals also require 30 molecules.

Supplementary Table S6. Paired differences against MiniMol and Chemprop (95% intervals).

Task	vs MiniMol	vs Chemprop
Bioavailability	+0.0255 [−0.0294, +0.0802]	+0.1221 [+0.0040, +0.2204]
Caco-2	+0.0745 [−0.0309, +0.1652]	+0.0503 [−0.0310, +0.1046]
HIA	−0.0008 [−0.0095, +0.0069]	+0.0125 [−0.0010, +0.0354]
Lipophilicity	−0.0246 [−0.0517, +0.0039]	−0.0401 [−0.0668, −0.0147]
P-gp	−0.0059 [−0.0286, +0.0081]	+0.0337 [−0.0037, +0.0666]
BBB	+0.0023 [−0.0127, +0.0168]	+0.0303 [+0.0050, +0.0614]
PPBR	+0.5468 [−0.2630, +1.4051]	+0.5595 [+0.0617, +1.0508]
VDss	+0.1018 [+0.0243, +0.1842]	+0.1271 [+0.0495, +0.2159]
CYP2C9 sub.	−0.0266 [−0.1113, +0.0601]	+0.0430 [−0.0469, +0.1321]
CYP2C9 inh.	−0.0368 [−0.0561, −0.0176]	+0.0261 [+0.0040, +0.0465]
CYP2D6 sub.	+0.0429 [−0.0061, +0.1002]	+0.1394 [+0.0438, +0.2162]
CYP2D6 inh.	−0.0022 [−0.0288, +0.0312]	+0.0463 [+0.0217, +0.0757]
CYP3A4 sub.	+0.0266 [−0.0301, +0.0812]	+0.0750 [+0.0188, +0.1325]
CYP3A4 inh.	+0.0059 [−0.0046, +0.0172]	+0.0158 [+0.0041, +0.0291]
Hepatocyte CL	+0.0705 [−0.0085, +0.1575]	+0.2452 [+0.1457, +0.3606]
Microsome CL	+0.0170 [−0.0520, +0.0901]	+0.1059 [+0.0495, +0.2079]
Half-life	−0.0194 [−0.1451, +0.1118]	+0.0432 [−0.0483, +0.1328]
AMES	+0.0158 [−0.0034, +0.0343]	+0.0483 [+0.0320, +0.0645]
DILI	−0.0415 [−0.0844, −0.0034]	+0.0178 [−0.0226, +0.0610]
hERG	+0.0202 [−0.0416, +0.0755]	+0.0611 [−0.0201, +0.1495]
LD50	−0.0144 [−0.0476, +0.0136]	+0.0085 [−0.0262, +0.0381]
Solubility	+0.0006 [−0.0336, +0.0363]	+0.0264 [+0.0029, +0.0528]

All 22 tasks are listed. Across the complete paired-prediction analysis, intervals favor Trimole-Hybrid over Chemprop on 13 tasks and Chemprop on 1; the corresponding counts for MiniMol are 1 and 2. The 20-task count used in the main report excludes BBB, evaluated from a stored ensemble rather than the mean of five AUROCs, and solubility, evaluated on the matched-row sample. In that subset, the counts favoring Trimole-Hybrid are 11 versus Chemprop and 1 versus MiniMol. Full scores and input hashes accompany the table.

Whole-task uncertainty: shared-pool AutoML

Supplementary Table S7. Task-wise selection versus FLAML (120 s): paired differences on identical inputs.

Task	Task-wise selection	FLAML 120 s	Difference [95% interval]
Bioavailability	0.7228	0.6593	+0.0635 [−0.0427, +0.1665]
Caco-2	0.2824	0.3378	+0.0554 [+0.0063, +0.0902]
HIA	0.9940	0.9834	+0.0106 [+0.0008, +0.0269]
Lipophilicity	0.4729	0.4687	−0.0042 [−0.0155, +0.0073]
P-gp	0.9384	0.9304	+0.0079 [+0.0006, +0.0168]
BBB	0.9264	0.8862	+0.0403 [+0.0198, +0.0642]
PPBR	7.3306	8.6949	+1.3643 [+0.9082, +1.8727]
VDss	0.6610	0.4954	+0.1656 [+0.1163, +0.2177]
CYP2C9 sub.	0.4396	0.3837	+0.0559 [−0.0190, +0.1404]
CYP2C9 inh.	0.7898	0.7751	+0.0147 [−0.0009, +0.0302]
CYP2D6 sub.	0.7475	0.6492	+0.0984 [+0.0348, +0.1434]
CYP2D6 inh.	0.7192	0.7235	−0.0043 [−0.0095, +0.0013]
CYP3A4 sub.	0.6735	0.6743	−0.0008 [−0.0167, +0.0156]
CYP3A4 inh.	0.8841	0.8839	+0.0002 [−0.0009, +0.0014]
Hepatocyte CL	0.5379	0.5279	+0.0100 [−0.0062, +0.0246]
Microsome CL	0.6428	0.5611	+0.0817 [−0.0021, +0.1452]
Half-life	0.4914	0.3764	+0.1150 [+0.0476, +0.1789]
AMES	0.8719	0.8633	+0.0086 [+0.0015, +0.0152]
DILI	0.9086	0.8869	+0.0217 [−0.0153, +0.0575]
hERG	0.8638	0.8056	+0.0582 [+0.0024, +0.1114]
LD50	0.6032	0.6153	+0.0122 [−0.0154, +0.0375]
Solubility	0.7384	0.8209	+0.0825 [+0.0339, +0.1581]

This table uses the shared candidate inputs in Note 3B and the bootstrap protocol in Table S6. Intervals favor the task-wise configuration on 11 tasks and FLAML on none. The reconstructed task-wise arm keeps the 20 original configuration choices (Table S2); FLAML is fitted and selected on development data. Its scores are reported separately from the main-table reference. Paired differences for the other controls accompany the data.

Within-task molecular identity

Thirteen historical test rows share molecular connectivity with rows in the same task's train/validation set: AMES 3, BBB 1, CYP2C9 inhibition 1, CYP2D6 inhibition 1, CYP3A4 inhibition 1 and hERG 6. Three of the 13 also have matching full InChIKeys. Molecular identities were compared without using labels. We did not measure how these matches affect scores. Cross-task source overlap is reported in S12, and the full split-pair records are provided in the companion data.

Scale-free ablations and global configuration

These ablations use original benchmark predictions. For CYP3A4 substrate and hepatocyte clearance, the Full arm uses the related-task models in Table S13; main Table 1 and Table S3 use target-only replacements. The original component matrix is available under the ablation analysis in the GitHub supplementary data index. We repeated candidate selection after each pool restriction using the same ranking rule as the full-pool single-candidate reference. Relative change is $100 d_t (S_{\text{variant}} - S_{\text{ref}})/|S_{\text{ref}}|$, with $d_t = -1$ for MAE and $+1$ otherwise. Negative values indicate worse performance. The changes include pool restriction and reselection.

Supplementary Table S8. Ablation changes and MLP late-fusion scores across 22 tasks.

Task	Reference	Relative change (%)					MLP late fusion	
	score	No chem.	No EPT/3D	Single seed	Global	Full	Mean	SD
Bioavailability	0.644297	-15.28	0.00	0.24	0.40	12.10	0.6264	0.0140
Caco-2	0.365308	0.10	10.63	3.62	0.00	22.35	0.5107	0.0192
HIA	0.950864	-4.65	0.00	-0.55	2.02	4.57	0.9585	0.0042
Lipophilicity	0.557236	0.00	-0.76	0.00	-7.74	15.14	2.2488	0.0271
P-gp	0.929286	-13.89	0.18	-0.21	0.00	0.98	0.9074	0.0039
BBB	0.904808	-16.33	1.45	-0.26	0.00	2.54	0.8186	0.0054
PPBR	9.195011	-8.24	3.62	0.83	0.00	20.28	18.0304	1.8241
VDss	0.457086	-68.32	0.00	4.98	-8.19	44.61	0.1951	0.0882
CYP2C9 sub.	0.431753	-27.10	1.18	4.62	0.00	2.69	0.4291	0.0046
CYP2C9 inh.	0.771022	-30.02	0.00	-0.12	-2.54	2.54	0.6493	0.0251
CYP2D6 sub.	0.702106	-30.08	0.00	-0.51	-1.79	6.47	0.7473	0.0150
CYP2D6 inh.	0.696138	-49.39	0.00	0.67	-1.12	3.31	0.5901	0.0196
CYP3A4 sub.	0.670547	-9.57	0.00	0.44	-0.33	8.16	0.6915	0.0184
CYP3A4 inh.	0.861292	-23.72	0.00	0.12	-0.16	2.65	0.7476	0.0115
Hepatocyte CL	0.200320	26.78	128.97	-10.73	0.00	175.58	0.1955	0.0758
Microsome CL	0.438244	0.00	0.00	-0.01	35.87	46.68	-0.0620	0.1934
Half-life	0.401874	-17.45	0.00	4.42	-4.17	22.27	0.0006	0.0478
AMES	0.855241	-18.63	0.00	-0.02	-1.17	1.95	0.8388	0.0051
DILI	0.866609	-11.59	0.00	-1.01	4.23	4.85	0.8363	0.0162
hERG	0.816495	-9.93	-0.31	-1.42	0.00	5.74	0.7662	0.0077
LD50	0.619278	-13.74	-1.50	-0.41	0.00	2.62	0.7345	0.0211
Solubility	0.838202	-10.88	-3.39	0.26	0.00	11.90	1.0989	0.0851

Chemical-feature removal lowers performance on 18/22 tasks. By metric, the median change [IQR] (linearly interpolated percentiles, R type 7) is −12.7% [−15.5%, −9.8%] for the 8 AUROC tasks, −30.0% [−30.1%, −27.1%] for the 5 AUPRC tasks, −8.7% [−30.2%, +6.7%] for the 4 Spearman tasks and −8.2% [−10.9%, 0.0%] for the 5 MAE tasks; metrics are not pooled. EPT removal changes 10 selected candidates, with 6 improvements and 4 losses. The global configuration worsens 9/13 changed tasks. The single-seed control uses seed 101 of the reference. Full combines different configuration choices with prediction averaging; the table gives each task's change. MLP columns give raw task scores (metrics in S3), as five-run means and sample SDs ($n-1$ denominator). The one-hidden-layer tanh MLP used validation-ranked saved predictions, with input standardization and standardized regression targets. MLP fitting used validation predictions. Its input pool includes the related-task predictors for hepatocyte clearance and CYP3A4 substrate. The solubility MLP used 1,997 test rows; S3 uses 1,995 matched rows.

Supplementary Table S9. Validation ranks for the global model-family choice.

Family (C: ChemBERTa; K: KPGT; E: EPT)	Mean rank
FP+C+K+E+XGB	1.818
FP+C+K+XGB	1.955
FP+xgb	3.455
K+E+linear	5.591
C+K+E+linear	6.000
K+linear	6.000
C+K+linear	6.045
E+linear	6.364
C+linear	7.773

Validation ranks select FP+C+K+E XGBoost, agreeing with the global choice based on mean validation performance. The companion data include the raw historical restrictions and naive-MLP results.

Component-level OOF and official-test comparisons

S10: Twenty eligible development tasks, nine fixed candidates and 27 methods. DILI lacked the required component groups, and HIA lacked class support under the fixed split. OOF methods add component-fold training predictions and computation. S is regularized stacking; the suffix cf identifies additional component-level OOF meta-training predictions. L adapts a historical rule. F learns convex weights using a stabilized anchor and a relative-loss penalty; N removes that penalty, and A uses the anchor alone. W fits global convex weights regularized toward top-three averaging; G uses input-dependent convex weights from a linear softmax gate. B selects one candidate by pooled development performance; U averages all nine candidates. O uses grouped mean-utility top-3 ranking; T3 uses pooled ranking. C-prefixed rows are individual candidate families: deep denotes encoder features with SGD classification or Ridge regression heads; chem_fp denotes Morgan-fingerprint XGBoost models, optionally including the named encoder features. F and L are development-study controls; Table S2 defines the main benchmark configurations. The full method catalog, 540 scores, protocols and contrasts are in the companion data.

Supplementary Table S10. Mean ranks of 27 methods across 20 development tasks.

Method	Rank	Method	Rank
S_cf	6.400	C_chem_fp_chemberta_kpgt_ept_xgb	12.725
L	7.975	O_cf	12.950
N	8.375	T3_pooled	13.250
F	8.825	T3_pooled_cf	13.625
G_cf	9.550	C_chem_fp_chemberta_kpgt_xgb	14.275
S	9.950	U	14.675
W	9.975	C_chem_fp_xgb	18.725
W_cf	9.975	C_deep_kpgt_ridge_linear	21.925
G	10.225	C_deep_chemberta_kpgt_ept_ridge_linear	22.275
B_cf	10.750	C_deep_ept_ridge_linear	22.850
B	10.775	C_deep_chemberta_kpgt_ridge_linear	23.350
A	11.125	C_deep_kpgt_ept_ridge_linear	23.425
AutoML_cf	12.275	C_deep_chemberta_ridge_linear	25.100
O	12.675		

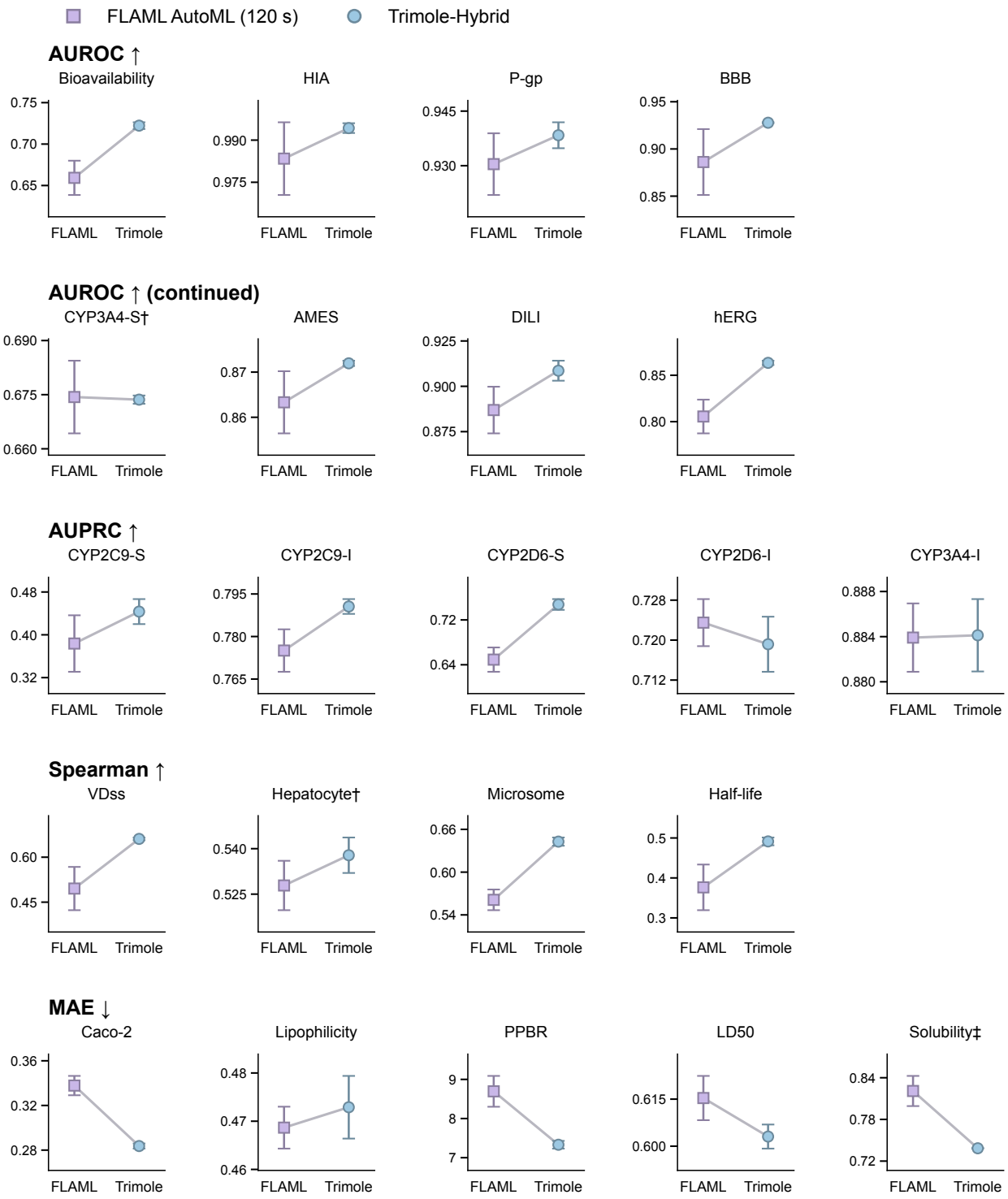
Component-level OOF stacking has the lowest mean rank (6.400). L ranks ahead of F overall, while F has better task means on 11 tasks and L on 9. The training regimes, including additional OOF fitting, are specified in Note 3B. AutoML_cf searched RF/ExtraTrees/XGBoost with at most 32 trials and a requested 60-s budget; S4 uses the expanded FLAML search setup.

Supplementary Table S11. Seven-method comparison on six official test tasks (five-run mean ± population SD).

Task	Metric	F	S	T3	N	A	O	L
BBB	AUROC	0.8866 ± 0.0110	0.8914 ± 0.0095	0.8856 ± 0.0128	0.8866 ± 0.0110	0.8875 ± 0.0120	0.8856 ± 0.0128	0.8886 ± 0.0128
Bioavailability	AUROC	0.6648 ± 0.0218	0.6749 ± 0.0297	0.6616 ± 0.0297	0.6654 ± 0.0224	0.6666 ± 0.0219	0.6591 ± 0.0259	0.6670 ± 0.0321
Hepatocyte CL	Spearman	0.4441 ± 0.0362	0.4175 ± 0.0575	0.4390 ± 0.0301	0.4434 ± 0.0369	0.4475 ± 0.0332	0.4435 ± 0.0296	0.4144 ± 0.0517
Microsome CL	Spearman	0.5782 ± 0.0307	0.5760 ± 0.0311	0.5352 ± 0.0514	0.5781 ± 0.0307	0.5632 ± 0.0411	0.5350 ± 0.0514	0.5478 ± 0.0334
CYP2C9 inh.	AUPRC	0.7704 ± 0.0049	0.7711 ± 0.0060	0.7620 ± 0.0029	0.7704 ± 0.0049	0.7620 ± 0.0029	0.7620 ± 0.0029	0.7593 ± 0.0031
CYP2D6 sub.	AUPRC	0.6773 ± 0.0253	0.6022 ± 0.0859	0.6748 ± 0.0245	0.6778 ± 0.0266	0.6778 ± 0.0257	0.6748 ± 0.0245	0.6608 ± 0.0418

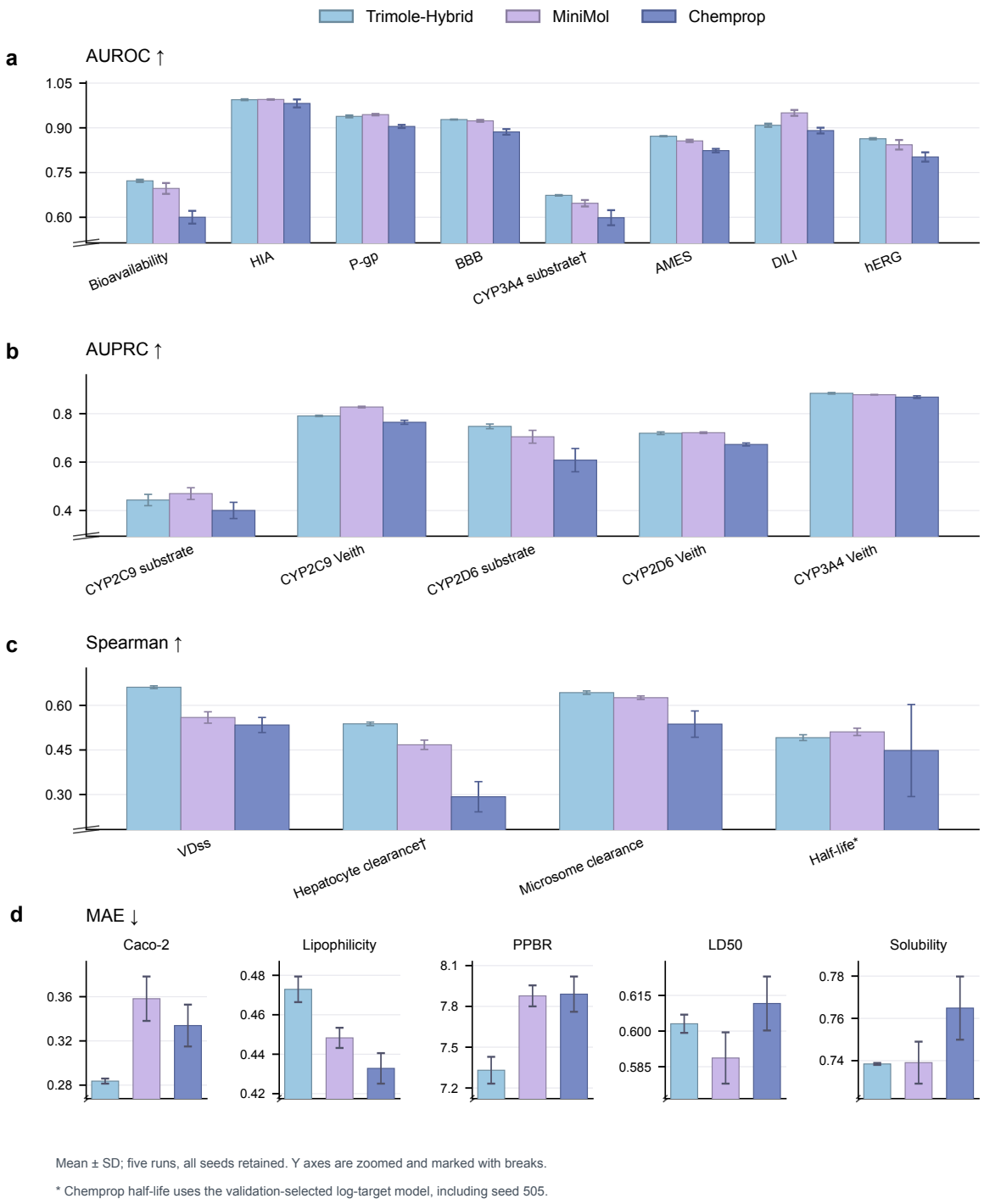
The seven methods and six-task subset were fixed before scoring. This study evaluates the configurations in Note 3B; the target-only models are reported in S13.

Prediction-level AutoML comparison: graphical summary



Supplementary Figure S1. FLAML (120 s) and the Trimole-Hybrid reference across 22 tasks. Error bars show five-run SDs. Against FLAML at 120 s, both the rebuilt task-wise configurations (S4) and the main-table reference plotted here have the better task mean on 19 of 22 tasks. FLAML searched for 120 s per task and run (protocol in Note 3B and Table S4). † marks a target-only Trimole-Hybrid replacement; ‡ marks solubility, for which both FLAML and the archived main-table reference cover 1,995 test rows.

External-model comparison: graphical summary



Mean ± SD; five runs, all seeds retained. Y axes are zoomed and marked with breaks.

* Chemprop half-life uses the validation-selected log-target model, including seed 505.

Supplementary Figure S2. Trimole-Hybrid, MiniMol and Chemprop across 22 tasks, grouped by official metric. Bars show five-run means and SDs; MAE panels keep task-specific units. The system reference includes two target-only replacements. Chemprop uses the validation-selected HIA checkpoints and half-life model. Training procedures are described in S3 and S5 and the MiniMol fit records. † marks a target-only Trimole-Hybrid replacement; the asterisk identifies the validation-selected log-target Chemprop half-life model, with all five runs included. Solubility uses the same 1,995 test rows for all three methods. The data-use procedures are given in Note 3A–B.

Related-task supervision and target-only models

Supplementary Table S12. Source tasks for the original related-task models.

Target endpoint	Related supervised source	Overlapping source rows
Hepatocyte clearance	Microsome clearance	109
CYP3A4 substrate	CYP2C9 / CYP2D6 substrate	162

Overlap counts refer to source-task rows whose molecules also occur in the target test set. These rows provide supervision for another endpoint, creating molecular overlap across task partitions. Source splits and molecular-match definitions are supplied with the data. The target-only models below use no related-task supervised rows and start from public pretrained representations.

CFA denotes Combinatorial Fusion Analysis, the method providing the archived public top-1 references in Table S13 and Figure S3.

Supplementary Table S13. Results for the two target-only models.

Task / metric	Related-task	Target-only mean ± SD	Public CFA
Hepatocyte / Spearman	0.552040	0.537834 ± 0.005849	0.536 ± 0.020
CYP3A4 sub. / AUROC	0.725249	0.673621 ± 0.001149	0.667 ± 0.019

Target-only models use supervised labels from the target endpoint, starting from public pretrained representations. Hepatocyte clearance used 970 development rows, three-fold identity-grouped validation and final development-data refitting. Its model components, blend weights and training settings are documented in the target-only training protocol in the GitHub supplementary data. CYP3A4 substrate used 535 development rows, three-fold scaffold validation and rank averaging of three RBF-SVMs on frozen C/K features. Imputation, scaling and PCA were fitted within the corresponding training folds. Five-run SDs use the population convention.

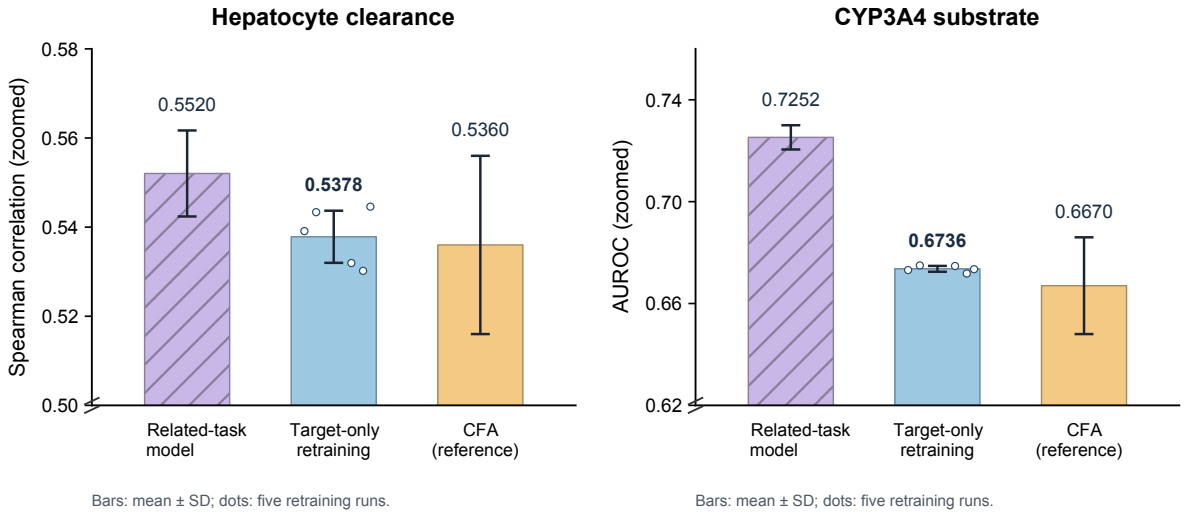


Figure S3. Related-task and target-only model results against the archived public top-1 reference (CFA). Dots show the five target-only runs; error bars show SD. CFA bars give the public mean; its SD is in Table S13. Within each training run, development validation selected the settings. Tables S12–S13 describe source-task supervision and target-only training. Molecular matches within task partitions are summarized on page 11.

Molecular-subgroup results

Subgroups use molecular weight (MW, 300 Da), cLogP (3) and topological polar surface area (TPSA, 75 Å^2). T/M/C: Trimole-Hybrid/MiniMol/Chemprop. Scores use matched test rows and the metrics in S3. Positive differences favor T (MAE reversed); Pos.: class prevalence (%).

Supplementary Table S14. Molecular-subgroup scores and paired differences across 22 tasks.

Task	Bin	n	Clusters	Pos.	T	M	C	Δ(T,M) [95% CI]	Δ(T,C) [95% CI]
Bioavailability	MW<300	31	24	87.1	0.5963	0.6167	0.7463	−0.0204 [NA]	−0.1500 [NA]
	MW≥300	97	72	72.2	0.7329	0.6940	0.5628	+0.0389 [−0.0212, +0.1023]	+0.1702 [+0.0566, +0.2746]
	cLogP<3	69	50	72.5	0.6796	0.6825	0.5515	−0.0029 [−0.0722, +0.0751]	+0.1281 [−0.0117, +0.2589]
	cLogP≥3	59	47	79.7	0.7691	0.6965	0.6543	+0.0727 [−0.0282, +0.2320]	+0.1149 [−0.0565, +0.2608]
	TPSA<75	74	58	82.4	0.6620	0.6525	0.6280	+0.0096 [−0.0775, +0.1120]	+0.0340 [−0.1089, +0.1574]
	TPSA≥75	54	38	66.7	0.7512	0.7093	0.5785	+0.0420 [−0.0210, +0.1191]	+0.1727 [+0.0341, +0.3256]
Caco-2	MW<300	34	18	–	0.2856	0.2900	0.3577	+0.0043 [NA]	+0.0720 [NA]
	MW≥300	148	60	–	0.2832	0.3739	0.3284	+0.0907 [−0.0447, +0.1924]	+0.0452 [−0.0640, +0.1164]
	cLogP<3	121	41	–	0.2889	0.4003	0.3639	+0.1114 [−0.0450, +0.2200]	+0.0750 [−0.0442, +0.1370]
	cLogP≥3	61	34	–	0.2733	0.2747	0.2743	+0.0014 [−0.0575, +0.0586]	+0.0011 [−0.0498, +0.0546]
	TPSA<75	43	28	–	0.2848	0.2749	0.3084	−0.0100 [NA]	+0.0236 [NA]
	TPSA≥75	139	50	–	0.2833	0.3840	0.3418	+0.1007 [−0.0439, +0.2001]	+0.0585 [−0.0522, +0.1266]
HIA	MW<300	50	46	76.0	0.9857	0.9925	0.9741	−0.0068 [−0.0324, +0.0113]	+0.0116 [−0.0029, +0.0324]
	MW≥300	67	43	77.6	1.0000	0.9997	0.9851	+0.0003 [+0.0000, +0.0017]	+0.0149 [+0.0000, +0.0632]
	cLogP<3	88	63	70.5	0.9940	0.9935	0.9789	+0.0005 [−0.0090, +0.0109]	+0.0151 [−0.0003, +0.0424]
	cLogP≥3	29	24	96.6	1.0000	1.0000	1.0000	+0.0000 [NA]	+0.0000 [NA]
	TPSA<75	57	49	91.2	1.0000	1.0000	1.0000	+0.0000 [+0.0000, +0.0000]	+0.0000 [+0.0000, +0.0000]
	TPSA≥75	60	41	63.3	0.9888	0.9911	0.9739	−0.0024 [−0.0215, +0.0144]	+0.0148 [−0.0063, +0.0534]
Lipophilicity	MW<300	159	108	–	0.5104	0.5082	0.4474	−0.0022 [−0.0690, +0.0712]	−0.0629 [−0.1245, −0.0021]
	MW≥300	681	387	–	0.4641	0.4343	0.4294	−0.0298 [−0.0598, +0.0008]	−0.0347 [−0.0625, −0.0068]
	cLogP<3	332	218	–	0.4880	0.4584	0.4198	−0.0297 [−0.0713, +0.0104]	−0.0682 [−0.1087, −0.0285]
	cLogP≥3	508	292	–	0.4630	0.4417	0.4413	−0.0213 [−0.0572, +0.0138]	−0.0217 [−0.0554, +0.0163]
	TPSA<75	304	206	–	0.4850	0.4734	0.4674	−0.0116 [−0.0612, +0.0368]	−0.0176 [−0.0652, +0.0334]
	TPSA≥75	536	294	–	0.4660	0.4340	0.4132	−0.0320 [−0.0632, +0.0013]	−0.0528 [−0.0809, −0.0236]
P-gp	MW<300	68	54	16.2	0.8128	0.8290	0.8179	−0.0163 [−0.1953, +0.0700]	−0.0051 [−0.1309, +0.0836]
	MW≥300	177	106	63.8	0.9377	0.9469	0.8908	−0.0092 [−0.0315, +0.0082]	+0.0469 [−0.0000, +0.0880]
	cLogP<3	114	87	13.2	0.8059	0.8246	0.8187	−0.0187 [−0.1307, +0.0587]	−0.0128 [−0.1121, +0.0561]
	cLogP≥3	131	70	83.2	0.8923	0.9098	0.7642	−0.0174 [−0.0629, +0.0178]	+0.1281 [+0.0238, +0.2473]
	TPSA<75	158	103	57.6	0.9514	0.9578	0.9214	−0.0065 [−0.0275, +0.0065]	+0.0300 [−0.0026, +0.0674]
	TPSA≥75	87	61	37.9	0.8994	0.9061	0.8561	−0.0066 [−0.0704, +0.0352]	+0.0433 [−0.0536, +0.0971]
BBB	MW<300	231	171	92.6	0.9170	0.8933	0.8862	+0.0236 [−0.0138, +0.0625]	+0.0308 [−0.0122, +0.0729]
	MW≥300	175	128	65.1	0.9169	0.9222	0.8985	−0.0053 [−0.0252, +0.0120]	+0.0184 [−0.0074, +0.0511]
	cLogP<3	275	203	80.7	0.9527	0.9442	0.9398	+0.0086 [−0.0077, +0.0273]	+0.0129 [−0.0045, +0.0330]
	cLogP≥3	131	95	80.9	0.8842	0.9000	0.8525	−0.0158 [−0.0475, +0.0123]	+0.0317 [−0.0168, +0.0855]
	TPSA<75	281	215	91.5	0.8857	0.8820	0.8079	+0.0037 [−0.0313, +0.0367]	+0.0778 [+0.0101, +0.1615]
	TPSA≥75	125	85	56.8	0.9142	0.9011	0.8978	+0.0130 [−0.0111, +0.0466]	+0.0164 [−0.0193, +0.0560]

Intervals: 2,000 paired scaffold bootstrap resamples, unadjusted. NA: insufficient support (S6; reasons in companion CSV). BBB uses stored ensembles; other scores average run metrics. Solubility: 1,995 matched rows.

Molecular-subgroup results (continued)

Subgroups use molecular weight (MW, 300 Da), cLogP (3) and topological polar surface area (TPSA, 75 Å^2). T/M/C: Trimole-Hybrid/MiniMol/Chemprop. Scores use matched test rows and the metrics in S3. Positive differences favor T (MAE reversed); Pos.: class prevalence (%).

Supplementary Table S14 (continued). Molecular-subgroup scores and paired differences across 22 tasks.

Task	Bin	n	Clusters	Pos.	T	M	C	Δ(T,M) [95% CI]	Δ(T,C) [95% CI]
PPBR	MW<300	47	29	–	8.1733	12.0094	9.7294	+3.8361 [NA]	+1.5561 [NA]
	MW≥300	512	197	–	7.2533	7.4981	7.7213	+0.2448 [−0.6196, +1.2191]	+0.4680 [−0.0542, +0.9595]
	cLogP<3	165	82	–	11.1423	11.1049	12.4665	−0.0374 [−2.1047, +2.1571]	+1.3242 [−0.0927, +2.6069]
	cLogP≥3	394	148	–	5.7344	6.5258	5.9736	+0.7914 [+0.0403, +1.7245]	+0.2393 [−0.1661, +0.5988]
	TPSA<75	244	105	–	7.3880	8.1490	7.8402	+0.7610 [−0.4760, +2.0926]	+0.4522 [−0.2239, +1.0534]
	TPSA≥75	315	124	–	7.2861	7.6669	7.9288	+0.3808 [−0.6935, +1.5746]	+0.6427 [−0.0793, +1.3787]
VDss	MW<300	84	73	–	0.6258	0.5875	0.5118	+0.0383 [−0.0973, +0.1758]	+0.1140 [−0.0059, +0.2609]
	MW≥300	142	119	–	0.6706	0.6278	0.5395	+0.0428 [−0.0537, +0.1433]	+0.1312 [+0.0261, +0.2409]
	cLogP<3	183	160	–	0.6546	0.5256	0.5772	+0.1290 [+0.0487, +0.2086]	+0.0774 [−0.0079, +0.1654]
	cLogP≥3	43	34	–	0.6014	0.6891	0.2610	−0.0877 [−0.3348, +0.1743]	+0.3404 [+0.0673, +0.6731]
	TPSA<75	75	65	–	0.6745	0.6377	0.6396	+0.0368 [−0.0554, +0.1330]	+0.0349 [−0.0810, +0.1393]
	TPSA≥75	151	127	–	0.6024	0.4759	0.4080	+0.1266 [+0.0141, +0.2379]	+0.1944 [+0.0711, +0.3296]
CYP2C9 sub.	MW<300	39	33	33.3	0.5423	0.4644	0.4892	+0.0779 [−0.0772, +0.2489]	+0.0531 [−0.1473, +0.2514]
	MW≥300	96	77	26.0	0.4170	0.4899	0.3780	−0.0729 [−0.1576, +0.0180]	+0.0389 [−0.0501, +0.1362]
	cLogP<3	57	48	22.8	0.4032	0.4274	0.3640	−0.0241 [−0.1824, +0.1381]	+0.0392 [−0.1139, +0.2151]
	cLogP≥3	78	60	32.1	0.5060	0.5134	0.4574	−0.0074 [−0.1017, +0.0934]	+0.0486 [−0.0625, +0.1573]
	TPSA<75	91	70	28.6	0.4628	0.4752	0.4296	−0.0124 [−0.1057, +0.0898]	+0.0333 [−0.0689, +0.1464]
	TPSA≥75	44	41	27.3	0.4559	0.5026	0.3980	−0.0467 [−0.1693, +0.0911]	+0.0579 [−0.0712, +0.1949]
CYP2C9 inh.	MW<300	773	579	11.9	0.5954	0.6517	0.5417	−0.0563 [−0.1152, +0.0070]	+0.0537 [−0.0205, +0.1311]
	MW≥300	1646	1180	41.2	0.8103	0.8462	0.7857	−0.0359 [−0.0558, −0.0139]	+0.0245 [+0.0022, +0.0475]
	cLogP<3	1213	892	15.1	0.5612	0.6270	0.5899	−0.0659 [−0.1174, −0.0172]	−0.0288 [−0.0662, +0.0096]
	cLogP≥3	1206	864	48.7	0.8347	0.8703	0.8033	−0.0356 [−0.0567, −0.0144]	+0.0314 [+0.0085, +0.0553]
	TPSA<75	1427	989	33.6	0.7860	0.8209	0.7614	−0.0349 [−0.0598, −0.0098]	+0.0247 [−0.0027, +0.0522]
	TPSA≥75	992	780	29.3	0.8007	0.8369	0.7701	−0.0362 [−0.0656, −0.0068]	+0.0306 [+0.0017, +0.0609]
CYP2D6 sub.	MW<300	54	40	33.3	0.8113	0.7685	0.8005	+0.0429 [−0.0381, +0.1400]	+0.0108 [−0.1619, +0.1401]
	MW≥300	81	64	30.9	0.7230	0.6948	0.5138	+0.0281 [−0.0318, +0.0859]	+0.2092 [+0.0972, +0.2823]
	cLogP<3	61	46	26.2	0.7193	0.6706	0.6358	+0.0487 [−0.0503, +0.1152]	+0.0835 [−0.0334, +0.1884]
	cLogP≥3	74	56	36.5	0.7697	0.7328	0.6121	+0.0370 [−0.0205, +0.1110]	+0.1576 [+0.0266, +0.2457]
	TPSA<75	91	68	37.4	0.8096	0.7636	0.6668	+0.0460 [−0.0125, +0.1069]	+0.1429 [+0.0330, +0.2422]
	TPSA≥75	44	35	20.5	0.5861	0.5766	0.3785	+0.0095 [−0.2506, +0.1653]	+0.2076 [−0.0052, +0.3383]
CYP2D6 inh.	MW<300	1091	537	10.7	0.5643	0.5747	0.4954	−0.0103 [−0.0691, +0.0734]	+0.0689 [+0.0108, +0.1416]
	MW≥300	1535	931	21.2	0.7733	0.7685	0.7373	+0.0048 [−0.0222, +0.0333]	+0.0360 [+0.0129, +0.0621]
	cLogP<3	1580	807	9.1	0.5096	0.5691	0.4751	−0.0595 [−0.1155, +0.0267]	+0.0345 [−0.0208, +0.0960]
	cLogP≥3	1046	657	28.7	0.8036	0.7852	0.7624	+0.0184 [−0.0090, +0.0464]	+0.0412 [+0.0180, +0.0652]
	TPSA<75	1484	811	22.5	0.7663	0.7574	0.7054	+0.0088 [−0.0192, +0.0407]	+0.0608 [+0.0352, +0.0892]
	TPSA≥75	1142	685	9.5	0.5612	0.5944	0.5560	−0.0332 [−0.0901, +0.0427]	+0.0052 [−0.0456, +0.0626]

Intervals: 2,000 paired scaffold bootstrap resamples, unadjusted. NA: insufficient support (S6; reasons in companion CSV). BBB uses stored ensembles; other scores average run metrics. Solubility: 1,995 matched rows.

Molecular-subgroup results (continued)

Subgroups use molecular weight (MW, 300 Da), cLogP (3) and topological polar surface area (TPSA, 75 Å^2). T/M/C: Trimole-Hybrid/MiniMol/Chemprop. Scores use matched test rows and the metrics in S3. Positive differences favor T (MAE reversed); Pos.: class prevalence (%).

Supplementary Table S14 (continued). Molecular-subgroup scores and paired differences across 22 tasks.

Task	Bin	n	Clusters	Pos.	T	M	C	Δ(T,M) [95% CI]	Δ(T,C) [95% CI]
CYP3A4 sub.	MW<300	45	33	60.0	0.6126	0.6115	0.5893	+0.0010 [−0.0862, +0.1045]	+0.0233 [−0.0556, +0.1177]
	MW≥300	90	72	57.8	0.7100	0.6758	0.6301	+0.0342 [−0.0340, +0.1060]	+0.0799 [+0.0195, +0.1416]
	cLogP<3	52	38	51.9	0.7876	0.7333	0.6367	+0.0542 [−0.0141, +0.1398]	+0.1508 [+0.0796, +0.2351]
	cLogP≥3	83	63	62.7	0.5781	0.5603	0.5550	+0.0178 [−0.0781, +0.1209]	+0.0231 [−0.0569, +0.1123]
	TPSA<75	94	69	60.6	0.6651	0.6538	0.5820	+0.0114 [−0.0613, +0.0861]	+0.0832 [+0.0072, +0.1652]
	TPSA≥75	41	36	53.7	0.7182	0.6675	0.6388	+0.0507 [−0.0269, +0.1453]	+0.0794 [+0.0144, +0.1492]
CYP3A4 inh.	MW<300	580	399	23.1	0.8178	0.8105	0.8234	+0.0073 [−0.0348, +0.0515]	−0.0056 [−0.0409, +0.0262]
	MW≥300	1887	1297	50.2	0.8908	0.8860	0.8740	+0.0048 [−0.0066, +0.0169]	+0.0168 [+0.0038, +0.0310]
	cLogP<3	1101	719	25.0	0.7754	0.7818	0.7921	−0.0064 [−0.0372, +0.0267]	−0.0167 [−0.0464, +0.0124]
	cLogP≥3	1366	964	59.1	0.9118	0.9034	0.8913	+0.0084 [−0.0023, +0.0203]	+0.0205 [+0.0083, +0.0341]
	TPSA<75	1456	963	44.2	0.8957	0.8869	0.8802	+0.0088 [−0.0042, +0.0230]	+0.0155 [+0.0011, +0.0316]
	TPSA≥75	1011	742	43.3	0.8694	0.8674	0.8559	+0.0020 [−0.0161, +0.0198]	+0.0135 [−0.0038, +0.0325]
Hepatocyte CL	MW<300	64	35	–	0.5960	0.5102	0.4445	+0.0858 [−0.0608, +0.3616]	+0.1515 [+0.0106, +0.2817]
	MW≥300	179	114	–	0.5271	0.4490	0.2314	+0.0781 [−0.0220, +0.1799]	+0.2957 [+0.1505, +0.4246]
	cLogP<3	105	63	–	0.4370	0.2839	0.2529	+0.1530 [+0.0466, +0.3144]	+0.1840 [+0.0551, +0.3293]
	cLogP≥3	138	88	–	0.6088	0.6311	0.3261	−0.0223 [−0.1507, +0.0913]	+0.2827 [+0.1291, +0.4425]
	TPSA<75	110	59	–	0.6081	0.5346	0.3998	+0.0735 [−0.0537, +0.2509]	+0.2083 [+0.1065, +0.3536]
	TPSA≥75	133	90	–	0.4963	0.4303	0.2203	+0.0660 [−0.0475, +0.1754]	+0.2760 [+0.1156, +0.4471]
Microsome CL	MW<300	31	24	–	0.3071	0.4435	0.1777	−0.1364 [NA]	+0.1294 [NA]
	MW≥300	190	102	–	0.6912	0.6481	0.5837	+0.0431 [−0.0220, +0.1270]	+0.1074 [+0.0477, +0.2173]
	cLogP<3	55	42	–	0.3499	0.4238	0.3119	−0.0738 [−0.2809, +0.0636]	+0.0380 [−0.1442, +0.2491]
	cLogP≥3	166	86	–	0.7142	0.6656	0.6034	+0.0486 [−0.0203, +0.1487]	+0.1108 [+0.0572, +0.2233]
	TPSA<75	104	56	–	0.6108	0.6185	0.5224	−0.0077 [−0.1361, +0.1314]	+0.0884 [−0.0004, +0.1976]
	TPSA≥75	117	75	–	0.6689	0.6291	0.5513	+0.0398 [−0.0314, +0.1035]	+0.1176 [+0.0436, +0.2568]
Half-life	MW<300	43	34	–	0.3053	0.3761	0.3865	−0.0708 [−0.4156, +0.2640]	−0.0813 [−0.3436, +0.1238]
	MW≥300	92	74	–	0.5547	0.5647	0.4694	−0.0100 [−0.1425, +0.1389]	+0.0854 [−0.0163, +0.1895]
	cLogP<3	82	70	–	0.4279	0.4354	0.4236	−0.0076 [−0.1944, +0.1763]	+0.0043 [−0.1391, +0.1336]
	cLogP≥3	53	37	–	0.5176	0.4505	0.3776	+0.0671 [−0.1804, +0.3308]	+0.1399 [−0.0237, +0.2824]
	TPSA<75	62	48	–	0.3076	0.3646	0.2768	−0.0570 [−0.3466, +0.2200]	+0.0308 [−0.1628, +0.2124]
	TPSA≥75	73	60	–	0.5791	0.6161	0.5437	−0.0370 [−0.1738, +0.1100]	+0.0354 [−0.0770, +0.1395]
AMES	MW<300	1035	277	63.8	0.8717	0.8536	0.8266	+0.0181 [−0.0071, +0.0410]	+0.0450 [+0.0252, +0.0646]
	MW≥300	422	220	49.8	0.8599	0.8501	0.8056	+0.0098 [−0.0206, +0.0397]	+0.0543 [+0.0240, +0.0892]
	cLogP<3	811	254	59.4	0.8608	0.8365	0.8157	+0.0243 [−0.0120, +0.0584]	+0.0452 [+0.0233, +0.0683]
	cLogP≥3	646	230	60.1	0.8848	0.8769	0.8336	+0.0079 [−0.0112, +0.0246]	+0.0512 [+0.0320, +0.0726]
	TPSA<75	1021	319	59.6	0.8804	0.8591	0.8303	+0.0213 [−0.0010, +0.0444]	+0.0501 [+0.0333, +0.0678]
	TPSA≥75	436	173	59.9	0.8505	0.8497	0.8054	+0.0008 [−0.0349, +0.0312]	+0.0450 [+0.0083, +0.0802]

Intervals: 2,000 paired scaffold bootstrap resamples, unadjusted. NA: insufficient support (S6; reasons in companion CSV). BBB uses stored ensembles; other scores average run metrics. Solubility: 1,995 matched rows.

Molecular-subgroup results (continued)

Subgroups use molecular weight (MW, 300 Da), cLogP (3) and topological polar surface area (TPSA, 75 Å^2). T/M/C: Trimole-Hybrid/MiniMol/Chemprop. Scores use matched test rows and the metrics in S3. Positive differences favor T (MAE reversed); Pos.: class prevalence (%).

Supplementary Table S14 (continued). Molecular-subgroup scores and paired differences across 22 tasks.

Task	Bin	n	Clusters	Pos.	T	M	C	Δ(T,M) [95% CI]	Δ(T,C) [95% CI]
DILI	MW<300	34	28	58.8	0.9171	0.9771	0.9050	−0.0600 [NA]	+0.0121 [NA]
	MW≥300	62	52	48.4	0.9254	0.9446	0.9052	−0.0192 [−0.0599, +0.0110]	+0.0202 [−0.0238, +0.0669]
	cLogP<3	50	42	52.0	0.9244	0.9801	0.8837	−0.0558 [−0.1309, −0.0062]	+0.0407 [−0.0165, +0.0986]
	cLogP≥3	46	39	52.2	0.8841	0.9186	0.8970	−0.0345 [−0.1129, +0.0404]	−0.0129 [−0.0699, +0.0357]
	TPSA<75	60	50	41.7	0.8969	0.9369	0.8517	−0.0400 [−0.1011, +0.0133]	+0.0453 [−0.0015, +0.1046]
	TPSA≥75	36	32	69.4	0.9156	0.9607	0.9273	−0.0451 [−0.1510, +0.0068]	−0.0116 [−0.1231, +0.0674]
hERG	MW<300	28	20	46.4	0.8118	0.8426	0.7667	−0.0308 [NA]	+0.0451 [NA]
	MW≥300	104	54	80.8	0.8360	0.7976	0.7809	+0.0384 [−0.0548, +0.1326]	+0.0551 [−0.0520, +0.1672]
	cLogP<3	42	37	45.2	0.8357	0.8714	0.7538	−0.0357 [−0.1276, +0.0409]	+0.0819 [−0.0445, +0.2137]
	cLogP≥3	90	34	86.7	0.7890	0.7237	0.6831	+0.0653 [−0.0640, +0.1774]	+0.1059 [−0.1348, +0.2920]
	TPSA<75	97	43	82.5	0.8324	0.8144	0.7124	+0.0180 [−0.0918, +0.0900]	+0.1201 [−0.0360, +0.2741]
	TPSA≥75	35	30	48.6	0.8804	0.8850	0.8242	−0.0046 [−0.0857, +0.0828]	+0.0562 [−0.0374, +0.1590]
LD50	MW<300	927	267	–	0.5465	0.5316	0.5503	−0.0149 [−0.0388, +0.0079]	+0.0038 [−0.0184, +0.0261]
	MW≥300	551	246	–	0.6982	0.6847	0.7147	−0.0135 [−0.0700, +0.0340]	+0.0165 [−0.0507, +0.0785]
	cLogP<3	850	277	–	0.5156	0.5063	0.5296	−0.0093 [−0.0298, +0.0089]	+0.0139 [−0.0040, +0.0331]
	cLogP≥3	628	235	–	0.7214	0.7002	0.7226	−0.0213 [−0.0750, +0.0311]	+0.0012 [−0.0652, +0.0683]
	TPSA<75	1234	369	–	0.6047	0.5907	0.6112	−0.0141 [−0.0498, +0.0182]	+0.0065 [−0.0308, +0.0409]
	TPSA≥75	244	142	–	0.5947	0.5786	0.6136	−0.0161 [−0.0541, +0.0224]	+0.0189 [−0.0196, +0.0590]
Solubility	MW<300	1172	326	–	0.6489	0.6443	0.6893	−0.0046 [−0.0477, +0.0353]	+0.0404 [+0.0155, +0.0691]
	MW≥300	823	356	–	0.8660	0.8740	0.8725	+0.0080 [−0.0468, +0.0653]	+0.0065 [−0.0358, +0.0535]
	cLogP<3	1217	407	–	0.6968	0.6915	0.7197	−0.0053 [−0.0418, +0.0293]	+0.0229 [−0.0051, +0.0503]
	cLogP≥3	778	275	–	0.8036	0.8134	0.8355	+0.0098 [−0.0472, +0.0714]	+0.0319 [−0.0182, +0.0933]
	TPSA<75	1279	391	–	0.7117	0.7001	0.7286	−0.0116 [−0.0572, +0.0310]	+0.0169 [−0.0136, +0.0518]
	TPSA≥75	716	302	–	0.7862	0.8087	0.8297	+0.0225 [−0.0316, +0.0800]	+0.0435 [−0.0000, +0.0832]

Intervals: 2,000 paired scaffold bootstrap resamples, unadjusted. NA: insufficient support (S6; reasons in companion CSV). BBB uses stored ensembles; other scores average run metrics. Solubility: 1,995 matched rows.